

## 4.1 Linear Functionals and the Dual Space

Chapter 3 studied linear maps  $V \rightarrow W$  into arbitrary targets. The smallest non-zero target is the field  $F$  itself, and maps into it are everywhere: a coordinate, the value of a polynomial at a point, an integral, the trace. This section names these maps and shows that they form a vector space  $V^*$ , of the same dimension as  $V$  when  $V$  is finite-dimensional. It also proves the fact that makes them useful: every non-zero vector is detected by at least one of them.

### 4.1.1 Measuring a vector by a number

After studying vector spaces and the maps between them, we look at the simplest possible target. Here are four maps we have already met, from four different families.

- **A coordinate.** Fix a basis  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  of  $V$ . The map  $V \rightarrow F$  sending  $\mathbf{v}$  to the first entry of  $[\mathbf{v}]_{\mathcal{B}}$  respects sums and multiples, by Theorem 1.5.2.
- **Evaluation.** The map  $F[x] \rightarrow F$ ,  $p \mapsto p(3)$ , satisfies  $(p + q)(3) = p(3) + q(3)$  and  $(cp)(3) = cp(3)$ , by Theorem 0.8.5 (a constant  $c$  is a polynomial with value  $c$ ).
- **Integration.** The map  $\mathbb{R}[x] \rightarrow \mathbb{R}$ ,  $p \mapsto \int_0^1 p(t) dt$ , is linear by the linearity of the integral from calculus.
- **Trace.** The map  $\text{tr}: M_n(F) \rightarrow F$  is linear (Example (Trace and Transpose)).

The inputs are columns, polynomials and matrices. The outputs are always single scalars. Each map is a way of **measuring** a vector by one number, and each measurement respects sums and scalar multiples. The same expression keeps recurring, “a linear map whose target is  $F$ ”, and we will see that these maps form a space with a structure of its own. So we give them a name.

*A linear functional is a linear measurement: it assigns a scalar to each vector, and the measurement of a linear combination is the same combination of the measurements.*

#### Definition 4.1.1 — Linear Functional

Let  $V$  be a vector space over  $F$ , and regard  $F$  as a vector space over itself. A **linear functional** on  $V$  is a linear map  $\varphi: V \rightarrow F$ . That is, for all  $\mathbf{u}, \mathbf{v} \in V$  and all  $c \in F$ ,

$$\varphi(\mathbf{u} + \mathbf{v}) = \varphi(\mathbf{u}) + \varphi(\mathbf{v}), \quad \varphi(c\mathbf{v}) = c\varphi(\mathbf{v}).$$

#### Definition 4.1.2 — Dual Space

Let  $V$  be a vector space over  $F$ . The **dual space** of  $V$  is the set of **all** linear functionals on  $V$ ,

$$V^* := \mathcal{L}(V, F),$$

with the pointwise operations  $(\varphi + \psi)(\mathbf{v}) := \varphi(\mathbf{v}) + \psi(\mathbf{v})$  and  $(c\varphi)(\mathbf{v}) := c\varphi(\mathbf{v})$  for  $\varphi, \psi \in V^*$ ,  $c \in F$  and  $\mathbf{v} \in V$ .

In words: a linear functional is a linear map (Definition 3.1.1) whose codomain is the one-dimensional space  $F$ . Its values lie in the **same** field  $F$  over which  $V$  is a vector space. The first condition says that measuring a sum gives the sum of the measurements, and the second says

that scaling a vector scales its measurement. The dual space collects **every** such measurement, and it adds them the natural way: the measurement  $\varphi + \psi$  reports the sum of the two readings. We write functionals with Greek letters  $\varphi, \psi$ , which keeps them apart from vectors of  $V$ .

**Well-definedness.** Two things need checking: that  $\varphi + \psi$  and  $c\varphi$  are again linear, and that the vector space axioms hold. Both are the case  $W = F$  of Theorem 3.4.1. So  $V^*$  is a vector space over  $F$ . Its zero vector is the **zero functional**  $\mathbf{v} \mapsto 0$ , and the negative of  $\varphi$  is  $\mathbf{v} \mapsto -\varphi(\mathbf{v})$ . Note that  $V^*$  is a set of **functions on  $V$** , not a subset of  $V$ .

### Example. (Four Families of Functionals)

Show that each map is a linear functional.

- (a)  $\varepsilon_c: F[x] \rightarrow F, p \mapsto p(c)$ , for a fixed  $c \in F$  (**evaluation at  $c$** ).
- (b)  $\iota: \mathbb{R}[x]_{\leq 2} \rightarrow \mathbb{R}, p \mapsto \int_0^1 p(t) dt$ .
- (c)  $\text{tr}: M_n(F) \rightarrow F$ .
- (d) The zero functional on any  $V$ . Also find the dual space of the zero space  $\{\mathbf{0}\}$ .

*Solution.*

- (a) By Theorem 0.8.5,  $(p+q)(c) = p(c) + q(c)$ , and, regarding  $ap$  as the product of the constant polynomial  $a$  with  $p$ ,  $(ap)(c) = a(c)p(c) = ap(c)$ , because the constant polynomial  $a$  has value  $a$  at  $c$ . These are the two conditions of Definition 4.1.1.
- (b) By linearity of the integral,  $\int_0^1 (p+q) = \int_0^1 p + \int_0^1 q$  and  $\int_0^1 ap = a \int_0^1 p$ . On coefficients,  $\iota(a_0 + a_1x + a_2x^2) = a_0 + \frac{a_1}{2} + \frac{a_2}{3}$ .
- (c) This is Theorem 0.9.6 (1):  $\text{tr}(\mathbf{A} + \mathbf{B}) = \text{tr} \mathbf{A} + \text{tr} \mathbf{B}$  and  $\text{tr}(c\mathbf{A}) = c \text{tr} \mathbf{A}$ .
- (d)  $0 = 0 + 0$  and  $0 = c \cdot 0$  in  $F$ , so the zero functional is linear. On  $\{\mathbf{0}\}$ , a linear functional must send  $\mathbf{0}$  to 0 (Theorem 3.1.1), and there is only one vector. So the zero functional is the only element, and  $\{\mathbf{0}\}^* = \{\mathbf{0}\}$ . This degenerate case matters: it is the case  $n = 0$  of the count  $\dim V^* = \dim V$  proved below.

The family  $F^n$  can be described completely. A row vector times a column is a scalar, and every functional on  $F^n$  arises this way.

### Theorem 4.1.1 — Functionals on $F^n$ Are Row Vectors

For  $\mathbf{a} = (a_1, \dots, a_n) \in F^n$ , define  $\varphi_{\mathbf{a}}: F^n \rightarrow F$  by

$$\varphi_{\mathbf{a}}(\mathbf{x}) := \mathbf{a}^\top \mathbf{x} = a_1x_1 + \dots + a_nx_n.$$

Then  $\varphi_{\mathbf{a}} \in (F^n)^*$ . Every  $\varphi \in (F^n)^*$  equals  $\varphi_{\mathbf{a}}$  for **exactly one  $\mathbf{a}$** , namely  $\mathbf{a} = (\varphi(\mathbf{e}_1), \dots, \varphi(\mathbf{e}_n))$ . The map  $F^n \rightarrow (F^n)^*, \mathbf{a} \mapsto \varphi_{\mathbf{a}}$ , is an isomorphism.

*Proof.* The map  $\varphi_{\mathbf{a}}$  is  $T_{\mathbf{a}^\top}$  for the  $1 \times n$  matrix  $\mathbf{a}^\top$ , with a  $1 \times 1$  matrix read as a scalar, so it is linear by Example (Multiplication by a Matrix). Let  $\varphi \in (F^n)^*$  and put  $a_i := \varphi(\mathbf{e}_i)$ . Then  $\varphi_{\mathbf{a}}(\mathbf{e}_i) = a_i = \varphi(\mathbf{e}_i)$  for each  $i$ . Two linear maps that agree on the basis  $(\mathbf{e}_1, \dots, \mathbf{e}_n)$  are equal by the uniqueness part of Theorem 3.1.5, so  $\varphi = \varphi_{\mathbf{a}}$ . If  $\varphi_{\mathbf{a}} = \varphi_{\mathbf{b}}$ , then  $a_i = \varphi_{\mathbf{a}}(\mathbf{e}_i) = \varphi_{\mathbf{b}}(\mathbf{e}_i) = b_i$  for every  $i$ , so  $\mathbf{a} = \mathbf{b}$ .

Hence  $\mathbf{a} \mapsto \varphi_{\mathbf{a}}$  is a bijection  $F^n \rightarrow (F^n)^*$ . It is linear, since  $\varphi_{\mathbf{a}+c\mathbf{b}}(\mathbf{x}) = \sum_i (a_i + cb_i)x_i =$

$\varphi_{\mathbf{a}}(\mathbf{x}) + c \varphi_{\mathbf{b}}(\mathbf{x})$  for every  $\mathbf{x}$ . By Theorem 3.5.1 it is an isomorphism. ■

So in  $F^n$  a functional is a row vector acting by matrix multiplication, and its  $1 \times n$  matrix in the standard bases is  $(\varphi(\mathbf{e}_1) \cdots \varphi(\mathbf{e}_n))$ . This is how we compute with functionals: rows act on columns.

**Non-examples by minimal change.** Each of the following changes one of the examples slightly. In each case something still works, and we name a condition that fails.

- On  $\mathbb{R}^2$ , the length  $(x, y) \mapsto \sqrt{x^2 + y^2}$  (Example (Length)). It sends  $\mathbf{0}$  to 0 and respects **positive** scalars. But with  $c = -1$  and  $\mathbf{v} = (1, 0)$  it gives 1, not  $-1$ , so the scaling condition fails.
- On  $\mathbb{R}[x]$ , squaring the evaluation at 0:  $p \mapsto p(0)^2$ . The outputs are still scalars. But for  $p = q = 1$  we get  $(1 + 1)^2 = 4 \neq 1 + 1$ , so additivity fails.
- On  $M_2(F)$ , replace the trace  $a + d$  of  $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$  by  $ad - bc$ . Then  $\mathbf{E}_{11} \mapsto 0$  and  $\mathbf{E}_{22} \mapsto 0$ , but  $\mathbf{E}_{11} + \mathbf{E}_{22} = \mathbf{I}_2 \mapsto 1$ , so additivity fails. The expression  $ad - bc$  is linear in each **row** separately, which is a different property.

**Why this definition.** Why single out the target  $F$ ? Because nothing is lost. A linear map  $T: V \rightarrow F^m$  is the same thing as  $m$  functionals: if  $\varphi_i(\mathbf{v})$  denotes the  $i$ -th entry of  $T\mathbf{v}$ , then each  $\varphi_i$  is linear and  $T\mathbf{v} = (\varphi_1(\mathbf{v}), \dots, \varphi_m(\mathbf{v}))$ ; conversely any  $m$  functionals stacked this way give a linear map. So functionals are the “rows” of linear maps, and understanding them one at a time is a way of understanding all maps into  $F^m$ . The name comes from analysis, where the vectors were themselves functions and a map such as  $f \mapsto \int_0^1 f$  was called a “functional”, a function of a function. The word “dual” will explain itself in this chapter: every statement about  $V$  has a mirror statement about  $V^*$ .

**Warning.** The values must lie in the field of scalars of  $V$ . Regard  $\mathbb{C}$  as a vector space over  $\mathbb{C}$ . The real part  $z \mapsto \operatorname{Re} z$  is additive, and it respects **real** scalars, but it is **not** a linear functional on  $\mathbb{C}$  over  $\mathbb{C}$ : with  $c = i$  and  $z = 1$ ,  $\operatorname{Re}(i \cdot 1) = 0 \neq i \cdot \operatorname{Re}(1) = i$ . It is a functional on  $\mathbb{C}$  regarded as a vector space over  $\mathbb{R}$ . The dual space depends on the field.

The first payoff is a count that holds for **every** non-zero functional: it hits every scalar, and in finite dimension it kills exactly one dimension.

### Proposition 4.1.2 — A Non-Zero Functional Is Surjective

Let  $V$  be a vector space over  $F$ , and let  $\varphi \in V^*$  be **non-zero**. Then  $\varphi$  is surjective. If  $V$  is finite-dimensional, then  $\dim \ker \varphi = \dim V - 1$ .

*Proof.* Since  $\varphi \neq 0$ , there is  $\mathbf{u} \in V$  with  $c := \varphi(\mathbf{u}) \neq 0$ . For every  $a \in F$ , linearity gives  $\varphi(ac^{-1}\mathbf{u}) = ac^{-1}c = a$ , so  $\varphi$  is surjective. Hence  $\operatorname{im} \varphi = F$  and  $\operatorname{rank} \varphi = 1$ . If  $V$  is finite-dimensional, the Rank–Nullity Theorem (Theorem 3.3.1) gives  $\dim \ker \varphi = \dim V - 1$ . ■

For example,  $\ker \operatorname{tr}$  in  $M_n(F)$  has dimension  $n^2 - 1$ , and the polynomials in  $F[x]_{\leq n}$  with  $p(c) = 0$  form a subspace of dimension  $n$ . Every non-zero functional cuts out a subspace one dimension

smaller than the whole space. Later in this chapter these kernels, the hyperplanes, become the building blocks of every subspace.

*Quick check.* Which of the following are linear functionals? (a)  $p \mapsto p(0) + 1$  on  $\mathbb{R}[x]$ . (b)  $p \mapsto p(1) - 2p'(0)$  on  $\mathbb{R}[x]$ . (c)  $\mathbf{A} \mapsto a_{12}$  on  $M_2(F)$ .

*Solution.*

- (a) No. It sends the zero polynomial to 1, but a linear map sends 0 to 0 (Theorem 3.1.1). (b) Yes. Evaluation at 1, differentiation and evaluation at 0 are linear, and  $V^*$  is closed under linear combinations and composition with linear maps (Theorem 3.4.1, Theorem 3.4.2). (c) Yes. It is the coordinate of  $\mathbf{A}$  at  $\mathbf{E}_{12}$  in the standard basis of Example (Standard Bases), and coordinates are linear (Theorem 1.5.2).

## 4.1.2 The dual basis

Return to the coordinate functionals. Let  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  be a basis of  $V$ , and let  $\varphi_i(\mathbf{v})$  be the  $i$ -th entry of  $[\mathbf{v}]_{\mathcal{B}}$ . The basis vector  $\mathbf{v}_j$  has coordinate vector  $\mathbf{e}_j$ , so

$$\varphi_i(\mathbf{v}_j) = \delta_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j, \end{cases}$$

with the Kronecker delta of Definition 0.9.3. Each  $\varphi_i$  reads off one coordinate and ignores the others. These  $n$  functionals are the natural candidates for a basis of  $V^*$ , and the theorem says they are one.

### Theorem 4.1.3 — The Dual Basis

Let  $V$  be a finite-dimensional vector space over  $F$ , and let  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  be a basis of  $V$ .

- (a) There are **unique**  $\varphi_1, \dots, \varphi_n \in V^*$  with  $\varphi_i(\mathbf{v}_j) = \delta_{ij}$  for all  $i, j$ . For every  $\mathbf{v} \in V$ ,  $\varphi_i(\mathbf{v})$  is the  $i$ -th entry of  $[\mathbf{v}]_{\mathcal{B}}$ .  
 (b) The list  $(\varphi_1, \dots, \varphi_n)$  is a basis of  $V^*$ .  
 (c) For every  $\mathbf{v} \in V$  and every  $\varphi \in V^*$ ,

$$\mathbf{v} = \sum_{i=1}^n \varphi_i(\mathbf{v}) \mathbf{v}_i, \quad \varphi = \sum_{i=1}^n \varphi(\mathbf{v}_i) \varphi_i.$$

*Idea.* ① A functional is a linear map into  $F$ , and a linear map may be prescribed freely on a basis, so existence and uniqueness in (a) are one citation. ② Independence: apply a vanishing combination  $\sum c_i \varphi_i$  to  $\mathbf{v}_j$ ; the delta kills every term but  $c_j$ . ③ Spanning: we cannot count, because  $\dim V^*$  is exactly what we do not know yet. So we guess the coefficients. If  $\varphi = \sum c_i \varphi_i$ , then applying both sides to  $\mathbf{v}_j$  forces  $c_j = \varphi(\mathbf{v}_j)$ . With this guess, the two sides are linear maps that agree on a basis, hence equal.

*Proof.*

- (a) For each  $i$ , apply Theorem 3.1.5 with  $W = F$  and the prescribed values  $\delta_{i1}, \dots, \delta_{in} \in F$ . It gives exactly one linear map  $\varphi_i: V \rightarrow F$  with  $\varphi_i(\mathbf{v}_j) = \delta_{ij}$  for all  $j$ . If  $\mathbf{v} = a_1\mathbf{v}_1 + \dots + a_n\mathbf{v}_n$ , then by Theorem 3.1.4

$$\varphi_i(\mathbf{v}) = \sum_{j=1}^n a_j \varphi_i(\mathbf{v}_j) = \sum_{j=1}^n a_j \delta_{ij} = a_i,$$

the  $i$ -th entry of  $[\mathbf{v}]_{\mathcal{B}}$  (Definition 1.5.2).

- (b) *Independence.* Let  $c_1\varphi_1 + \dots + c_n\varphi_n = 0$  in  $V^*$ . Applying both sides to  $\mathbf{v}_j$  and using the pointwise operations of Definition 4.1.2,

$$0 = \sum_{i=1}^n c_i \varphi_i(\mathbf{v}_j) = \sum_{i=1}^n c_i \delta_{ij} = c_j$$

for each  $j$ . Hence  $(\varphi_1, \dots, \varphi_n)$  is linearly independent.

*Spanning.* Let  $\varphi \in V^*$ , and put  $\psi := \sum_{i=1}^n \varphi(\mathbf{v}_i) \varphi_i \in V^*$ . For each  $j$ ,  $\psi(\mathbf{v}_j) = \sum_i \varphi(\mathbf{v}_i) \delta_{ij} = \varphi(\mathbf{v}_j)$ . So  $\psi$  and  $\varphi$  are linear maps  $V \rightarrow F$  that agree on the basis  $\mathcal{B}$ , and  $\psi = \varphi$  by the uniqueness part of Theorem 3.1.5. Hence  $\varphi \in \text{span}(\varphi_1, \dots, \varphi_n)$ , and  $(\varphi_1, \dots, \varphi_n)$  is a basis of  $V^*$ .

- (c) The second formula was proved in the spanning step. The first is the definition of coordinates, since  $\varphi_i(\mathbf{v})$  is the  $i$ -th coordinate of  $\mathbf{v}$  by (a). This proves the theorem. ■

The two formulas in (c) are mirror images. A vector is recovered from its measurements by the dual basis; a functional is recovered from its values on the basis.

### Definition 4.1.3 — Dual Basis

Let  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  be a basis of a finite-dimensional vector space  $V$  over  $F$ . The basis  $(\varphi_1, \dots, \varphi_n)$  of  $V^*$  given by Theorem 4.1.3, characterized by  $\varphi_i(\mathbf{v}_j) = \delta_{ij}$ , is **the dual basis** of  $\mathcal{B}$ . We write it  $\mathcal{B}^*$ .

Counting the dual basis gives the dimension of the dual space at once.

### Corollary 4.1.4 — Dimension of the Dual Space

Let  $V$  be a finite-dimensional vector space over  $F$ . Then  $V^*$  is finite-dimensional and  $\dim V^* = \dim V$ . In particular  $V^* \cong V$ .

*Proof.* Let  $n = \dim V$  and choose a basis of  $V$  (Corollary 1.5.5). By Theorem 4.1.3 (b), its dual basis is a basis of  $V^*$  of length  $n$ . Hence  $\dim V^* = n$ , and  $V^* \cong V$  by Theorem 3.5.4. ■

For instance  $\dim M_{m \times n}(F)^* = mn$  and  $\dim F[x]_{\leq n}^* = n + 1$ .

*Remark.* Many books write  $\mathbf{v}_i^*$  for the dual basis functional  $\varphi_i$ . We avoid that notation for two reasons. It suggests that  $\varphi_i$  is determined by  $\mathbf{v}_i$  alone, which is false (see the warning below). And in this book the star on a map is reserved for adjoints, which come much later. We write  $\varphi_i$ , and speak of “the dual basis of  $\mathcal{B}$ ”.

How do we find a dual basis in practice? In  $F^n$ , functionals are rows, and the conditions  $\varphi_i(\mathbf{v}_j) = \delta_{ij}$  turn into a single matrix equation.

**Example. (The Dual Basis Is Read Off the Inverse Matrix)**

- (a) Find the dual basis of  $\mathcal{B} = ((1, 1), (1, -1))$  in  $\mathbb{R}^2$ .
- (b) More generally, let  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  be a basis of  $F^n$ , and let  $\mathbf{P} = (\mathbf{v}_1 \ \cdots \ \mathbf{v}_n)$ . Show that  $\mathbf{P}$  is invertible and that the dual basis of  $\mathcal{B}$  is  $\varphi_i(\mathbf{x}) = \mathbf{r}_i \mathbf{x}$ , where  $\mathbf{r}_i$  is the  $i$ -th **row** of  $\mathbf{P}^{-1}$ .

*Solution.*

- (a) By [Theorem 4.1.1](#),  $\varphi_1(x, y) = px + qy$  for some  $p, q \in \mathbb{R}$ , that is,  $\varphi_1(\mathbf{x}) = (p \ q) \mathbf{x}$ . The conditions  $\varphi_1(1, 1) = 1$  and  $\varphi_1(1, -1) = 0$  say  $(p \ q) \mathbf{P} = (1 \ 0)$  with  $\mathbf{P} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ . Likewise the row  $\mathbf{r}_2$  of  $\varphi_2$  satisfies  $\mathbf{r}_2 \mathbf{P} = (0 \ 1)$ . Stacking the two rows into a matrix  $\mathbf{R}$ , the four conditions become  $\mathbf{R} \mathbf{P} = \mathbf{I}_2$ , so  $\mathbf{R} = \mathbf{P}^{-1}$ . By Theorem 0.9.8, with  $ad - bc = -2$ ,

$$\mathbf{P}^{-1} = \frac{1}{-2} \begin{pmatrix} -1 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix}.$$

Hence

$$\varphi_1(x, y) = \frac{x + y}{2}, \quad \varphi_2(x, y) = \frac{x - y}{2}.$$

Check:  $\varphi_1(1, 1) = 1$ ,  $\varphi_1(1, -1) = 0$ ,  $\varphi_2(1, 1) = 0$ ,  $\varphi_2(1, -1) = 1$ . And the first formula of [Theorem 4.1.3](#) (c) for  $\mathbf{v} = (3, 1)$  reads  $(3, 1) = 2(1, 1) + 1(1, -1)$ , which is true. The second formula expresses a functional in the dual basis through its values on  $\mathcal{B}$ : for  $\psi(x, y) = x$ , the values are  $\psi(1, 1) = 1$  and  $\psi(1, -1) = 1$ , so  $\psi = \varphi_1 + \varphi_2$ , and indeed  $\frac{x+y}{2} + \frac{x-y}{2} = x$ .

- (b) The columns of  $\mathbf{P}$  form a basis of  $F^n$ , so  $\mathbf{P}$  is invertible by Theorem 2.4.4. Let  $\mathbf{R} \in M_n(F)$  have rows  $\mathbf{r}_1, \dots, \mathbf{r}_n$ , and let  $\psi_i(\mathbf{x}) := \mathbf{r}_i \mathbf{x}$ . By the entry formula for a product (Theorem 0.9.2),  $\psi_i(\mathbf{v}_j) = \mathbf{r}_i \mathbf{v}_j = (\mathbf{R} \mathbf{P})_{ij}$ . Hence  $\psi_i(\mathbf{v}_j) = \delta_{ij}$  for all  $i, j$  if and only if  $\mathbf{R} \mathbf{P} = \mathbf{I}_n$ , if and only if  $\mathbf{R} = \mathbf{P}^{-1}$  (multiply by  $\mathbf{P}^{-1}$  on the right). Now take  $\mathbf{R} = \mathbf{P}^{-1}$ : the functionals  $\psi_i(\mathbf{x}) = \mathbf{r}_i \mathbf{x}$  are linear ([Theorem 4.1.1](#)) and satisfy  $\psi_i(\mathbf{v}_j) = \delta_{ij}$ , so by the uniqueness in [Theorem 4.1.3](#) (a) they are the dual basis.

This agrees with Chapter 3:  $\mathbf{P} = [\text{id}]_{\mathcal{B}}^{\mathcal{E}}$  and  $\mathbf{P}^{-1} = [\text{id}]_{\mathcal{E}}^{\mathcal{B}}$  (Theorem 3.7.1), so  $\mathbf{P}^{-1} \mathbf{x} = [\mathbf{x}]_{\mathcal{B}}$ , and row  $i$  of  $\mathbf{P}^{-1}$  computes the  $i$ -th coordinate, as [Theorem 4.1.3](#) (a) predicts. The columns of  $\mathbf{P}$  are the basis; the rows of  $\mathbf{P}^{-1}$  are the dual basis.

**Warning.** A dual basis functional depends on the whole basis, not only on its own vector. In  $\mathbb{R}^2$ , the basis  $((1, 0), (0, 1))$  has first dual functional  $\varphi_1(x, y) = x$ .

The basis  $((1, 0), (1, 1))$  has the same first vector, but  $\mathbf{P}^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$ , so its first dual functional is  $\varphi_1(x, y) = x - y$ . Changing  $\mathbf{v}_2$  changed  $\varphi_1$ , because  $\varphi_1$  must vanish on  $\mathbf{v}_2$ .

The same method works away from  $F^n$ , once we recognize the coordinates.

**Example. (Dual Bases in  $\mathbb{R}[x]_{\leq 2}$ )**

- (a) Show that the dual basis of  $(1, x, x^2)$  in  $\mathbb{R}[x]_{\leq 2}$  is  $(\varphi_0, \varphi_1, \varphi_2)$  with  $\varphi_0(p) = p(0)$ ,  $\varphi_1(p) = p'(0)$  and  $\varphi_2(p) = \frac{1}{2}p''(0)$ . (We index from 0 to match the exponents.)
- (b) Let  $\ell_{-1} = \frac{1}{2}x(x-1)$ ,  $\ell_0 = 1 - x^2$  and  $\ell_1 = \frac{1}{2}x(x+1)$ . Show that  $(\ell_{-1}, \ell_0, \ell_1)$  is a basis of  $\mathbb{R}[x]_{\leq 2}$  whose dual basis is  $(\varepsilon_{-1}, \varepsilon_0, \varepsilon_1)$ , the evaluations at  $-1, 0, 1$ . Deduce that  $p = p(-1)\ell_{-1} + p(0)\ell_0 + p(1)\ell_1$  for every  $p \in \mathbb{R}[x]_{\leq 2}$ .

*Solution.*

- (a) Each  $\varphi_k$  is linear, as a composition of differentiation (Example (Differentiation)) with evaluation at 0, scaled. For  $p = a_0 + a_1x + a_2x^2$ , we have  $p(0) = a_0$ ,  $p'(0) = a_1$  and  $\frac{1}{2}p''(0) = \frac{1}{2} \cdot 2a_2 = a_2$ . So  $\varphi_k(x^j) = \delta_{kj}$  for  $j, k \in \{0, 1, 2\}$ , and by the uniqueness in Theorem 4.1.3 (a) these are the dual basis. Over a general field  $F$ , the dual basis of  $(1, x, x^2)$  is still “the coefficient of  $x^k$ ”, but the formula  $\frac{1}{2}p''(0)$  needs  $2 \neq 0$  in  $F$ .
- (b) The values of the  $\ell$ ’s at the three points are

$$\begin{aligned}\ell_{-1}(-1) &= 1, \ell_{-1}(0) = 0, \ell_{-1}(1) = 0; \\ \ell_0(-1) &= 0, \ell_0(0) = 1, \ell_0(1) = 0; \\ \ell_1(-1) &= 0, \ell_1(0) = 0, \ell_1(1) = 1.\end{aligned}$$

That is,  $\varepsilon_i(\ell_j) = \delta_{ij}$  for  $i, j \in \{-1, 0, 1\}$ . *Basis.* Let  $c_{-1}\ell_{-1} + c_0\ell_0 + c_1\ell_1 = 0$ . Evaluating at  $j \in \{-1, 0, 1\}$  gives  $c_j = 0$ . So the list is independent, of length  $3 = \dim \mathbb{R}[x]_{\leq 2}$ , hence a basis by Theorem 1.6.4. *Dual basis.* The evaluations are linear (Example (Four Families of Functionals)) and satisfy  $\varepsilon_i(\ell_j) = \delta_{ij}$ , so they are the dual basis by the uniqueness in Theorem 4.1.3 (a). The first formula of Theorem 4.1.3 (c) now reads  $p = \sum_i \varepsilon_i(p) \ell_i = p(-1)\ell_{-1} + p(0)\ell_0 + p(1)\ell_1$ .

The last formula rebuilds a polynomial from three of its values. It is a preview of the Lagrange interpolation formula of Chapter 5, and it explains where the  $\ell$ ’s come from: they are the basis dual to the evaluations.

*Quick check.* Let  $(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$  be a basis of  $V$  with dual basis  $(\varphi_1, \varphi_2, \varphi_3)$ . Compute  $\varphi_2(3\mathbf{v}_1 - \mathbf{v}_2 + 5\mathbf{v}_3)$  and  $(2\varphi_1 + \varphi_3)(\mathbf{v}_3)$ .

*Solution.*  $\varphi_2(3\mathbf{v}_1 - \mathbf{v}_2 + 5\mathbf{v}_3) = 3 \cdot 0 - 1 \cdot 1 + 5 \cdot 0 = -1$ , the second coordinate.  $(2\varphi_1 + \varphi_3)(\mathbf{v}_3) = 2 \cdot 0 + 1 = 1$ .

**Warning.**  $V \cong V^*$  is true in finite dimension, but no isomorphism is singled out. Corollary 4.1.4 gives an isomorphism by sending  $\mathbf{v}_i \mapsto \varphi_i$ , and that depends on the basis. In  $\mathbb{R}^2$  with the standard basis, it sends  $(1, 0)$  to the functional  $(x, y) \mapsto x$ . With the basis  $((1, 0), (1, 1))$ , it sends the **same** vector  $(1, 0)$  to  $(x, y) \mapsto x - y$ , by the warning above. A vector does not come with “its” functional. Later in this chapter we make

precise the sense in which  $V^*$  needs a choice to be identified with  $V$ , while the double dual  $V^{**}$  does not.

### 4.1.3 Testing against functionals

In  $F^n$  a vector is zero exactly when all its entries are zero, and the entries are the values of the coordinate functionals. The same holds in any finite-dimensional space, and it gives a flexible way to prove that a vector is zero.

#### Theorem 4.1.5 — Functionals Separate Vectors

Let  $V$  be a finite-dimensional vector space over  $F$ .

- (a) If  $\mathbf{v} \in V$  and  $\mathbf{v} \neq \mathbf{0}$ , then there is  $\varphi \in V^*$  with  $\varphi(\mathbf{v}) = 1$ .
- (b) If  $\mathbf{v} \in V$  satisfies  $\varphi(\mathbf{v}) = 0$  for **every**  $\varphi \in V^*$ , then  $\mathbf{v} = \mathbf{0}$ . It suffices that  $\varphi_i(\mathbf{v}) = 0$  for the functionals  $\varphi_i$  of **one** dual basis.
- (c) For  $\mathbf{u}, \mathbf{w} \in V$ :  $\mathbf{u} = \mathbf{w}$  if and only if  $\varphi(\mathbf{u}) = \varphi(\mathbf{w})$  for every  $\varphi \in V^*$ .

*Idea.* We need a functional that does not vanish at  $\mathbf{v}$ , and the only functionals we can build at will are dual basis functionals. The first one takes the value 1 on the first basis vector. So make  $\mathbf{v}$  the first vector of a basis: this is possible exactly because  $\mathbf{v} \neq \mathbf{0}$ .

*Proof.*

- (a) Since  $\mathbf{v} \neq \mathbf{0}$ , the list  $(\mathbf{v})$  is linearly independent: if  $a\mathbf{v} = \mathbf{0}$ , then  $a = 0$  by Theorem 1.1.9. By the Basis Extension Theorem (Theorem 1.6.5), there is a basis  $(\mathbf{v}, \mathbf{v}_2, \dots, \mathbf{v}_n)$  of  $V$ . Let  $(\varphi_1, \dots, \varphi_n)$  be its dual basis (Theorem 4.1.3). Then  $\varphi_1(\mathbf{v}) = 1$ .
- (b) The first statement is the contrapositive of (a). For the second, if  $(\varphi_1, \dots, \varphi_n)$  is the dual basis of a basis  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  and every  $\varphi_i(\mathbf{v}) = 0$ , then  $\mathbf{v} = \sum_i \varphi_i(\mathbf{v})\mathbf{v}_i = \mathbf{0}$  by Theorem 4.1.3 (c).
- (c) If  $\mathbf{u} = \mathbf{w}$ , the values agree. Conversely, if  $\varphi(\mathbf{u}) = \varphi(\mathbf{w})$  for every  $\varphi$ , then  $\varphi(\mathbf{u} - \mathbf{w}) = \varphi(\mathbf{u}) - \varphi(\mathbf{w}) = 0$  for every  $\varphi$ , so  $\mathbf{u} - \mathbf{w} = \mathbf{0}$  by (b).

■

This is the signature move of the chapter: **to show a vector is zero, test it against every functional, and a dual basis suffices**. It turns one equation in  $V$  into scalar equations, which are often easier to reach.

*Remark.* Parts (a), (b) (first sentence) and (c) hold for **every** vector space  $V$ ; only (a) needs a new construction. By Theorem 1.8.2,  $\{\mathbf{v}\}$  extends to a basis  $B$  of  $V$ . Every  $\mathbf{w} \in V$  is a finite combination of distinct vectors of  $B$ , with coefficients that are unique because  $B$  is independent. Let  $\varphi(\mathbf{w})$  be the coefficient of  $\mathbf{v}$  in this expansion (and 0 if  $\mathbf{v}$  does not occur). Adding or scaling expansions gives expansions, so by uniqueness coefficients add and scale, and  $\varphi$  is linear with  $\varphi(\mathbf{v}) = 1$ . The price is Zorn's Lemma (Theorem 1.8.1): for  $\mathbb{R}$  over  $\mathbb{Q}$ , no one has an explicit formula for such a functional.

**Example. (A Matrix Killed by Every Trace)**

Let  $\mathbf{A} \in M_n(F)$  satisfy  $\text{tr}(\mathbf{A}\mathbf{X}) = 0$  for every  $\mathbf{X} \in M_n(F)$ . Show that  $\mathbf{A} = 0$ . Deduce that every linear functional on  $M_n(F)$  is  $\mathbf{X} \mapsto \text{tr}(\mathbf{A}\mathbf{X})$  for exactly one  $\mathbf{A} \in M_n(F)$ .

*Solution.* The matrix units  $\mathbf{E}_{ij}$  form a basis of  $M_n(F)$  (Example (Standard Bases)), and  $\mathbf{A} = \sum_{i,j} a_{ij} \mathbf{E}_{ij}$ . By Theorem 4.1.3 (a), its dual basis consists of the coordinate functionals  $\varepsilon_{ij}(\mathbf{A}) = a_{ij}$ . We compute  $\varepsilon_{ij}(\mathbf{A})$  as a trace. By Definition 0.9.9,  $(\mathbf{A}\mathbf{E}_{ji})_{kk} = \sum_l a_{kl}(\mathbf{E}_{ji})_{lk}$ , and  $(\mathbf{E}_{ji})_{lk} = 1$  only for  $l = j$  and  $k = i$ . So the only non-zero diagonal entry of  $\mathbf{A}\mathbf{E}_{ji}$  is  $(\mathbf{A}\mathbf{E}_{ji})_{ii} = a_{ij}$ , and

$$\text{tr}(\mathbf{A}\mathbf{E}_{ji}) = a_{ij} = \varepsilon_{ij}(\mathbf{A}).$$

Taking  $\mathbf{X} = \mathbf{E}_{ji}$  in the hypothesis gives  $\varepsilon_{ij}(\mathbf{A}) = 0$  for all  $i, j$ . By Theorem 4.1.5 (b),  $\mathbf{A} = 0$ .

For the deduction, let  $\Theta: M_n(F) \rightarrow M_n(F)^*$ ,  $\Theta(\mathbf{A})(\mathbf{X}) := \text{tr}(\mathbf{A}\mathbf{X})$ . Each  $\Theta(\mathbf{A})$  is linear in  $\mathbf{X}$ , and  $\Theta$  is linear in  $\mathbf{A}$ , both by Theorem 0.9.6 (1) and the distributive laws of Theorem 0.9.4. We just showed  $\ker \Theta = \{0\}$ , so  $\Theta$  is injective (Theorem 3.2.3). Since  $\dim M_n(F)^* = n^2 = \dim M_n(F)$  (Corollary 4.1.4),  $\Theta$  is bijective by Corollary 3.3.2 (e). This is the matrix analogue of Theorem 4.1.1.

## 4.1.4 When $V$ is infinite-dimensional

The main results of the last two subsections assumed finite dimension, or used a finite basis. That hypothesis does real work. Consider  $F[x]$ , with its basis  $\{1, x, x^2, \dots\}$  (Example (A Basis of  $F[x]$ )). For  $k \in \mathbb{N}$  let

$$\varphi_k(p) := \text{the coefficient of } x^k \text{ in } p.$$

These are the coordinate functionals, and  $\varphi_k(x^j) = \delta_{kj}$ , exactly as in Theorem 4.1.3. But they are not a basis of the dual space.

### Proposition 4.1.6 — The Coordinate Functionals Do Not Span $F[x]^*$

The functionals  $\varphi_k$  ( $k \in \mathbb{N}$ ) lie in  $F[x]^*$  and form a linearly independent set. But they do **not** span  $F[x]^*$ : the evaluation  $\varepsilon_1: p \mapsto p(1)$  is **not** a linear combination of finitely many  $\varphi_k$ .

*Proof.* Polynomials are added and scaled coefficient by coefficient (Definition 0.8.1), so each  $\varphi_k$  is linear. *Independence.* Let  $k_1, \dots, k_m$  be distinct and  $c_1\varphi_{k_1} + \dots + c_m\varphi_{k_m} = 0$ . Applying both sides to  $x^{k_j}$  gives  $c_j = 0$ , since  $\varphi_{k_i}(x^{k_j}) = \delta_{k_i k_j}$ .

*Not spanning.* The evaluation  $\varepsilon_1$  is linear by Example (Four Families of Functionals) (a), and  $\varepsilon_1(x^N) = 1^N = 1$  for every  $N \in \mathbb{N}$ . Suppose, for a contradiction, that  $\varepsilon_1 = c_1\varphi_{k_1} + \dots + c_m\varphi_{k_m}$  for some  $m \in \mathbb{N}$ , distinct  $k_j$  and  $c_j \in F$ . Choose  $N \in \mathbb{N}$  larger than every  $k_j$ . Then  $\varphi_{k_j}(x^N) = 0$  for every  $j$ , so the right-hand side sends  $x^N$  to 0, while  $\varepsilon_1(x^N) = 1$ . This contradiction shows  $\varepsilon_1 \notin \text{span}\{\varphi_k : k \in \mathbb{N}\}$ .

Where did the proof of [Theorem 4.1.3](#) break? In the spanning step. The candidate  $\sum_i \varphi(\mathbf{v}_i)\varphi_i$  for  $\varphi = \varepsilon_1$  would be  $\varphi_0 + \varphi_1 + \varphi_2 + \cdots$ , an infinite sum, and infinite sums mean nothing in a vector space (the warning in Chapter 1, §8). A functional on  $F[x]$  may take non-zero values on infinitely many basis vectors, while a finite combination of the  $\varphi_k$  cannot.

*Remark.* More is true, and we state it without proof. For every infinite-dimensional  $V$ , a basis of  $V^*$  is **strictly larger**, as an infinite set, than a basis of  $V$ ; in particular  $V^* \not\cong V$ . So “ $\dim V^* = \dim V$ ” is a genuinely finite-dimensional fact, and this is why finite dimension appears in the hypotheses of most theorems in this chapter. A first step towards seeing how large  $F[x]^*$  is appears in Exercise C4 below.

## 4.1.5 Exercises

### 4.1.5.1 A. Check your understanding

#### Exercise A1.

- (a) Define a linear functional on a vector space  $V$  over  $F$ , and the dual space  $V^*$ .
- (b) True or false:  $p \mapsto p(1)^2$  is a linear functional on  $\mathbb{R}[x]$ . Justify your answer.
- (c) True or false: if  $(\mathbf{v}_1, \mathbf{v}_2)$  and  $(\mathbf{v}_1, \mathbf{w}_2)$  are bases of  $V$ , then the first functionals of their dual bases are equal. Justify your answer.
- (d) What is  $\dim M_{2 \times 3}(F)^*$ ?
- (e) State the formula that expresses  $\varphi \in V^*$  in the dual basis of  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ .
- (f) Name the move you would use to prove that a vector  $\mathbf{v}$  of a finite-dimensional space is  $\mathbf{0}$  without computing  $\mathbf{v}$  itself.

*Solution.*

- (a) See [Definition 4.1.1](#) and [Definition 4.1.2](#): a linear functional is a linear map  $V \rightarrow F$ , and  $V^* = \mathcal{L}(V, F)$  with pointwise addition and scaling.
- (b) False. For  $p = 1$  and  $c = 2$ , the polynomial  $cp = 2$  is sent to  $2^2 = 4$ , while  $c$  times the image of  $p$  is  $2 \cdot 1^2 = 2$ . So the scaling condition fails.
- (c) False. By the warning after [Example \(The Dual Basis Is Read Off the Inverse Matrix\)](#), in  $\mathbb{R}^2$  the bases  $((1, 0), (0, 1))$  and  $((1, 0), (1, 1))$  have first dual functionals  $x$  and  $x - y$ .
- (d) 6, by [Corollary 4.1.4](#), since  $\dim M_{2 \times 3}(F) = 6$ .
- (e)  $\varphi = \sum_{i=1}^n \varphi(\mathbf{v}_i)\varphi_i$ , by [Theorem 4.1.3](#) (c).
- (f) Test against functionals: show  $\varphi(\mathbf{v}) = 0$  for every  $\varphi \in V^*$ , or only for the functionals of one dual basis, and apply [Theorem 4.1.5](#) (b).

### 4.1.5.2 B. Practice

**Exercise B1: Finding dual bases.**

- (a) Find the dual basis of  $((1, 0, 1), (0, 1, 1), (1, 1, 1))$  in  $\mathbb{R}^3$ .  
 (b) Show that the dual basis of  $(1, x - 1, (x - 1)^2)$  in  $\mathbb{R}[x]_{\leq 2}$  is  $p \mapsto p(1)$ ,  $p \mapsto p'(1)$ ,  $p \mapsto \frac{1}{2}p''(1)$ . Hence write  $p = 2 + 3x - x^2$  in this basis.

*Solution.*

- (a) Let  $\mathbf{P}$  have the three vectors as columns. Row reduction of  $[\mathbf{P} \mid \mathbf{I}_3]$  (Theorem 2.4.6) gives

$$\mathbf{P} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \quad \mathbf{P}^{-1} = \begin{pmatrix} 0 & -1 & 1 \\ -1 & 0 & 1 \\ 1 & 1 & -1 \end{pmatrix},$$

and one checks  $\mathbf{P}^{-1}\mathbf{P} = \mathbf{I}_3$ . By [Example \(The Dual Basis Is Read Off the Inverse Matrix\)](#) (b), the dual basis is

$$\varphi_1(x, y, z) = -y + z, \quad \varphi_2(x, y, z) = -x + z, \quad \varphi_3(x, y, z) = x + y - z.$$

For instance  $\varphi_3(1, 0, 1) = 0$ ,  $\varphi_3(0, 1, 1) = 0$  and  $\varphi_3(1, 1, 1) = 1$ .

- (b) The three maps are linear, as combinations of differentiation and evaluation at 1. Call them  $\psi_1, \psi_2, \psi_3$ . On the basis:

$$\begin{aligned} \psi_1(1) &= 1, \quad \psi_1(x - 1) = 0, \quad \psi_1((x - 1)^2) = 0; \\ \psi_2(1) &= 0, \quad \psi_2(x - 1) = 1, \quad \psi_2((x - 1)^2) = 2(1 - 1) = 0; \\ \psi_3(1) &= 0, \quad \psi_3(x - 1) = 0, \quad \psi_3((x - 1)^2) = \frac{1}{2} \cdot 2 = 1. \end{aligned}$$

So  $\psi_i$  takes the value  $\delta_{ij}$  on the  $j$ -th basis vector, and by the uniqueness in [Theorem 4.1.3](#) (a) they are the dual basis. (The list  $(1, x - 1, (x - 1)^2)$  is a basis: it has distinct degrees 0, 1, 2, so it is independent by Theorem 1.4.4, and it has length  $3 = \dim \mathbb{R}[x]_{\leq 2}$ , so Theorem 1.6.4 applies.) For  $p = 2 + 3x - x^2$ :  $p(1) = 4$ ,  $p'(1) = 3 - 2 = 1$ ,  $\frac{1}{2}p''(1) = -1$ . By [Theorem 4.1.3](#) (c),

$$p = 4 + (x - 1) - (x - 1)^2.$$

Check:  $4 + x - 1 - (x^2 - 2x + 1) = 2 + 3x - x^2$ .

**Exercise B2: Which maps are functionals?.**

Determine which of the following are linear functionals. Justify your answer.

- (a)  $\mathbb{R}^3 \rightarrow \mathbb{R}$ ,  $(x, y, z) \mapsto 2x - z$ .  
 (b)  $\mathbb{R}^3 \rightarrow \mathbb{R}$ ,  $(x, y, z) \mapsto xy$ .  
 (c)  $M_2(\mathbb{R}) \rightarrow \mathbb{R}$ ,  $\mathbf{A} \mapsto a_{12} - 3a_{21}$ .  
 (d)  $\mathbb{R}[x] \rightarrow \mathbb{R}$ ,  $p \mapsto \int_0^1 t p(t) dt$ .  
 (e)  $\mathbb{R}[x]_{\leq 2} \rightarrow \mathbb{R}$ ,  $p \mapsto p(0) + p'(0) + 1$ .  
 (f)  $\mathbb{C}^2 \rightarrow \mathbb{C}$ ,  $(z, w) \mapsto z + \overline{w}$ , where  $\mathbb{C}^2$  is a vector space over  $\mathbb{C}$ .

*Solution.*

- (a) Yes: it is  $\varphi_{\mathbf{a}}$  with  $\mathbf{a} = (2, 0, -1)$ , linear by [Theorem 4.1.1](#).
- (b) No.  $(1, 1, 0) \mapsto 1$ , but  $2(1, 1, 0) = (2, 2, 0) \mapsto 4 \neq 2$ , so the scaling condition fails.
- (c) Yes: it is  $\varepsilon_{12} - 3\varepsilon_{21}$ , a combination of coordinate functionals for the standard basis, and  $V^*$  is a vector space ([Definition 4.1.2](#)).
- (d) Yes. By linearity of the integral,  $\int_0^1 t(p+q)(t) dt = \int_0^1 tp(t) dt + \int_0^1 tq(t) dt$  and  $\int_0^1 t(cp)(t) dt = c \int_0^1 tp(t) dt$ .
- (e) No. The zero polynomial is sent to  $1 \neq 0$ , contradicting [Theorem 3.1.1](#).
- (f) No. With  $c = i$  and  $(z, w) = (0, 1)$ :  $(0, i) \mapsto \bar{i} = -i$ , while  $i \cdot (0 + \bar{1}) = i$ . The map is additive and respects real scalars, but it is not linear over  $\mathbb{C}$ .

### Exercise B3: Coordinates of a functional.

- (a) Let  $(\varphi_1, \varphi_2)$  be the dual basis of  $((1, 1), (1, -1))$  from [Example \(The Dual Basis Is Read Off the Inverse Matrix\)](#). Write  $\psi(x, y) = 3x + 5y$  as a combination of  $\varphi_1$  and  $\varphi_2$ .
- (b) Let  $(\varphi_0, \varphi_1, \varphi_2)$  be the dual basis of  $(1, x, x^2)$  from [Example \(Dual Bases in  \$\mathbb{R}\[x\]\_{\leq 2}\$ \)](#). Write  $\iota(p) = \int_0^1 p(t) dt$  as a combination of them.

*Solution.*

- (a) By [Theorem 4.1.3](#) (c),  $\psi = \psi(1, 1)\varphi_1 + \psi(1, -1)\varphi_2 = 8\varphi_1 - 2\varphi_2$ . Check:  $8 \cdot \frac{x+y}{2} - 2 \cdot \frac{x-y}{2} = 4x + 4y - x + y = 3x + 5y$ .
- (b) By [Theorem 4.1.3](#) (c),  $\iota = \iota(1)\varphi_0 + \iota(x)\varphi_1 + \iota(x^2)\varphi_2 = \varphi_0 + \frac{1}{2}\varphi_1 + \frac{1}{3}\varphi_2$ . Check: for  $p = a_0 + a_1x + a_2x^2$ , the right-hand side is  $a_0 + \frac{a_1}{2} + \frac{a_2}{3} = \int_0^1 p$ .

### 4.1.5.3 C. Going deeper

#### Exercise C1: Functionals with the same kernel.

Let  $V$  be **any** vector space over  $F$ , and let  $\varphi, \psi \in V^*$  with  $\ker \varphi = \ker \psi$ . Prove that  $\psi = c\varphi$  for some  $c \in F$ . Is the converse true?

*Hint: if  $\varphi \neq 0$ , fix  $\mathbf{u}$  with  $\varphi(\mathbf{u}) = 1$  and consider  $\mathbf{v} - \varphi(\mathbf{v})\mathbf{u}$ .*

*Solution.* If  $\varphi = 0$ , then  $\ker \psi = \ker \varphi = V$ , so  $\psi = 0 = 1 \cdot \varphi$ . Suppose  $\varphi \neq 0$ . By [Proposition 4.1.2](#) there is  $\mathbf{u} \in V$  with  $\varphi(\mathbf{u}) = 1$ . Let  $\mathbf{v} \in V$ . Then

$$\varphi(\mathbf{v} - \varphi(\mathbf{v})\mathbf{u}) = \varphi(\mathbf{v}) - \varphi(\mathbf{v})\varphi(\mathbf{u}) = 0,$$

so  $\mathbf{v} - \varphi(\mathbf{v})\mathbf{u} \in \ker \varphi = \ker \psi$ . Therefore  $0 = \psi(\mathbf{v} - \varphi(\mathbf{v})\mathbf{u})$ , that is,  $\psi(\mathbf{v}) = \psi(\mathbf{u})\varphi(\mathbf{v})$ . Since  $\mathbf{v}$  was arbitrary,  $\psi = c\varphi$  with  $c = \psi(\mathbf{u})$ .

The converse holds only for  $c \neq 0$ : if  $\psi = c\varphi$  with  $c \neq 0$ , then  $\psi(\mathbf{v}) = 0 \iff \varphi(\mathbf{v}) = 0$ . For  $c = 0$  and  $\varphi \neq 0$ ,  $\ker \psi = V \neq \ker \varphi$ .

**Exercise C2: Functionals vanishing where others vanish.**

Let  $V$  be a vector space over  $F$ , and let  $\varphi_1, \dots, \varphi_k, \psi \in V^*$ .

- (a) Prove that if  $\ker \varphi_1 \cap \dots \cap \ker \varphi_k \subseteq \ker \psi$ , then  $\psi \in \text{span}(\varphi_1, \dots, \varphi_k)$ .
- (b) Deduce that if  $V$  is finite-dimensional, then  $(\varphi_1, \dots, \varphi_k)$  spans  $V^*$  if and only if  $\ker \varphi_1 \cap \dots \cap \ker \varphi_k = \{\mathbf{0}\}$ .

*Hint: for (a), consider the linear map  $T: V \rightarrow F^k$ ,  $\mathbf{v} \mapsto (\varphi_1(\mathbf{v}), \dots, \varphi_k(\mathbf{v}))$ .*

*Solution.*

- (a) Let  $T$  be as in the hint; it is linear because each  $\varphi_i$  is. Its kernel is  $\ker \varphi_1 \cap \dots \cap \ker \varphi_k$ . Define  $S: \text{im } T \rightarrow F$  by  $S(T\mathbf{v}) := \psi(\mathbf{v})$ . This is well defined: if  $T\mathbf{v} = T\mathbf{v}'$ , then  $\mathbf{v} - \mathbf{v}' \in \ker T \subseteq \ker \psi$ , so  $\psi(\mathbf{v}) = \psi(\mathbf{v}')$ . It is linear, since  $S(cT\mathbf{v} + T\mathbf{v}') = S(T(c\mathbf{v} + \mathbf{v}')) = \psi(c\mathbf{v} + \mathbf{v}') = c\psi(\mathbf{v}) + \psi(\mathbf{v}') = cS(T\mathbf{v}) + S(T\mathbf{v}')$ . Now extend  $S$  to  $F^k$ . The subspace  $\text{im } T \subseteq F^k$  has a basis  $(\mathbf{y}_1, \dots, \mathbf{y}_r)$  (Theorem 1.6.7, Corollary 1.5.5), which extends to a basis  $(\mathbf{y}_1, \dots, \mathbf{y}_k)$  of  $F^k$  by Theorem 1.6.5. By Theorem 3.1.5 there is  $\tilde{S} \in (F^k)^*$  with  $\tilde{S}(\mathbf{y}_j) = S(\mathbf{y}_j)$  for  $j \leq r$  and  $\tilde{S}(\mathbf{y}_j) = 0$  for  $j > r$ . Then  $\tilde{S}$  and  $S$  agree on a basis of  $\text{im } T$ , so  $\tilde{S} = S$  on  $\text{im } T$ . By Theorem 4.1.1,  $\tilde{S}(z_1, \dots, z_k) = a_1 z_1 + \dots + a_k z_k$  for some  $a_i \in F$ . Hence for every  $\mathbf{v} \in V$ ,

$$\psi(\mathbf{v}) = S(T\mathbf{v}) = \tilde{S}(\varphi_1(\mathbf{v}), \dots, \varphi_k(\mathbf{v})) = a_1 \varphi_1(\mathbf{v}) + \dots + a_k \varphi_k(\mathbf{v}),$$

so  $\psi = a_1 \varphi_1 + \dots + a_k \varphi_k$ . Note that  $V$  was arbitrary; only  $F^k$  needed to be finite-dimensional.

- (b)  $(\Leftarrow)$  If the intersection is  $\{\mathbf{0}\}$ , it lies in  $\ker \psi$  for every  $\psi \in V^*$ , so every  $\psi$  is in the span by (a).  $(\Rightarrow)$  Suppose the  $\varphi_i$  span  $V^*$ , and let  $\mathbf{v}$  lie in every  $\ker \varphi_i$ . Every  $\psi \in V^*$  is a combination  $\sum c_i \varphi_i$ , so  $\psi(\mathbf{v}) = \sum c_i \varphi_i(\mathbf{v}) = 0$ . By Theorem 4.1.5 (b),  $\mathbf{v} = \mathbf{0}$ .

**Exercise C3: Evaluations as a dual basis.**

Let  $n \in \mathbb{N}$ .

- (a) Let  $x_0, \dots, x_n \in F$  be **distinct**. Prove that the evaluations  $(\varepsilon_{x_0}, \dots, \varepsilon_{x_n})$  form a basis of  $F[x]_{\leq n}^*$ , and that they are the dual basis of some basis of  $F[x]_{\leq n}$ .
- (b) Over  $\mathbb{F}_2$  with  $n = 2$ , show that the evaluations at the elements of  $\mathbb{F}_2$  do **not** span  $\mathbb{F}_2[x]_{\leq 2}^*$ .

*Hint: for (a), use Theorem 2.7.3 to build the basis first.*

*Solution.*

- (a) For each  $j$ , Theorem 2.7.3 gives  $\ell_j \in F[x]_{\leq n}$  with  $\ell_j(x_i) = \delta_{ij}$  for  $i = 0, \dots, n$ , because the  $x_i$  are distinct. If  $\sum_j c_j \ell_j = 0$ , evaluating at  $x_i$  gives  $c_i = 0$ . So  $(\ell_0, \dots, \ell_n)$  is independent of length  $n + 1 = \dim F[x]_{\leq n}$ , hence a basis by Theorem 1.6.4. The evaluations are linear (Example (Four Families of Functionals)) and  $\varepsilon_{x_i}(\ell_j) = \delta_{ij}$ , so by the uniqueness in Theorem 4.1.3 (a) they are the dual

basis of  $(\ell_0, \dots, \ell_n)$ , and by [Theorem 4.1.3](#) (b) a basis of  $F[x]_{\leq n}^*$ .

- (b)  $\mathbb{F}_2 = \{0, 1\}$  has only two elements, so only  $\varepsilon_0$  and  $\varepsilon_1$  are available, and  $\dim \mathbb{F}_2[x]_{\leq 2}^* = 3$  by [Corollary 4.1.4](#). Two functionals cannot span a 3-dimensional space ([Theorem 1.6.3](#)). Concretely,  $p = x^2 + x$  is non-zero but  $p(0) = 0$  and  $p(1) = 1 + 1 = 0$ ; the coefficient functional of  $x^2$  takes the value 1 on  $p$ , so it is not a combination of  $\varepsilon_0, \varepsilon_1$ . The hypothesis “distinct points” in (a) needs  $n + 1$  distinct elements of  $F$ , which a small field may not have.

#### Exercise C4: How large is $F[x]^*$ ?

Let  $F^{\mathbb{N}}$  be the space of sequences with entries in  $F$ .

- (a) Prove that  $\Phi: F[x]^* \rightarrow F^{\mathbb{N}}$ ,  $\varphi \mapsto (\varphi(1), \varphi(x), \varphi(x^2), \dots)$ , is an isomorphism.  
 (b) Identify  $\Phi(\varphi_k)$  for the coordinate functionals  $\varphi_k$  of [Proposition 4.1.6](#), and  $\Phi(\varepsilon_1)$ . Explain how [Proposition 4.1.6](#) matches Example (The Vectors  $\mathbf{e}_i$  Do Not Span  $F^{\mathbb{N}}$ ).

*Hint: for surjectivity, remember that a polynomial has only finitely many non-zero coefficients.*

*Solution.*

- (a) *Linear.* The  $k$ -th entry of  $\Phi(c\varphi + \psi)$  is  $(c\varphi + \psi)(x^k) = c\varphi(x^k) + \psi(x^k)$ , by the point-wise operations of [Definition 4.1.2](#). *Injective.* If  $\Phi(\varphi) = \mathbf{0}$ , then  $\varphi$  vanishes on the basis  $\{x^k\}$  of Example (A Basis of  $F[x]$ ). Every polynomial is a finite combination of the  $x^k$ , so  $\varphi = 0$  by [Theorem 3.1.4](#); hence  $\Phi$  is injective by [Theorem 3.2.3](#). *Surjective.* Given  $(a_0, a_1, \dots) \in F^{\mathbb{N}}$ , define  $\varphi(b_0 + b_1x + \dots + b_Nx^N) := a_0b_0 + \dots + a_Nb_N$ . This is a finite sum, it is well defined because coefficients are unique, and it is linear because coefficients add and scale; it is the linear map with  $\varphi(x^k) = a_k$  given by [Theorem 3.1.6](#). Then  $\Phi(\varphi) = (a_0, a_1, \dots)$ . By [Theorem 3.5.1](#),  $\Phi$  is an isomorphism.
- (b)  $\Phi(\varphi_k) = (\delta_{k0}, \delta_{k1}, \dots) = \mathbf{e}_k$ , and  $\Phi(\varepsilon_1) = (1, 1, 1, \dots)$ . An isomorphism carries spans to spans, so  $\varepsilon_1 \in \text{span}\{\varphi_k\}$  would give  $(1, 1, 1, \dots) \in \text{span}\{\mathbf{e}_k\}$ . Example (The Vectors  $\mathbf{e}_i$  Do Not Span  $F^{\mathbb{N}}$ ) showed this is false, because finite combinations of the  $\mathbf{e}_k$  have only finitely many non-zero entries. So  $F[x]$ , with its basis  $\{x^k\}$  indexed by  $\mathbb{N}$ , has a dual as large as the whole sequence space  $F^{\mathbb{N}}$ .